# Intelligent Hybrid Phishing Detection using URL Analysis, Vision Models, and LLM Reasoning

Dhruv Khandelwal (2022UCP1130)
Govind Ram Mali (2022UCP1630)

Department of Computer Science & Engineering
Malaviya National Institute of Technology Jaipur

**Supervisor:** Dr. Arka Prokash Mazumdar

May 2026

# Outline

# Introduction

- **Phishing** — fraudulent websites mimicking legitimate ones to steal sensitive information (credentials, financial data, personal details)
- Phishing attacks increased by **over 60%** in the past year, with millions of phishing sites detected monthly
- Traditional single-modality detection systems face critical limitations:
  - URL-based methods evaded via obfuscation, typosquatting, homograph attacks
  - Content-based methods fail against sophisticated visual mimicry
  - Lack of **interpretability** in detection decisions
- **Need:** A multi-modal, interpretable, and real-time detection system

# Problem Statement

## Core Challenge

Design a **robust, accurate, and interpretable** phishing detection system overcoming limitations of single-modality approaches.

The system must:

1. Analyze **multiple characteristics** — URL structure, visual appearance, semantic content
2. Provide **real-time detection** with sub-second latency
3. Generate **human-readable explanations** for decisions
4. Achieve **high accuracy** with low false positive rates
5. **Adapt** to evolving phishing techniques via modern ML/DL architectures

# Literature Review

| Approach | Key Work | Limitations |
|---|---|---|
| URL-Based | Ma et al. (2009), URLNet (2018) | Cannot detect visual mimicry; susceptible to obfuscation |
| Visual-Based | GoldPhish (2010), PhishPedia (2021) | Misses URL indicators; computationally expensive |
| Content-Based | CANTINA (2007), DeltaPhish (2017) | Requires page access; evaded by dynamic content |
| Blacklist | PhishNet (2010) | Cannot detect zero-day attacks |
| Multi-Modal | Marchal et al. (2017) | Lacks interpretability; no LLM integration |

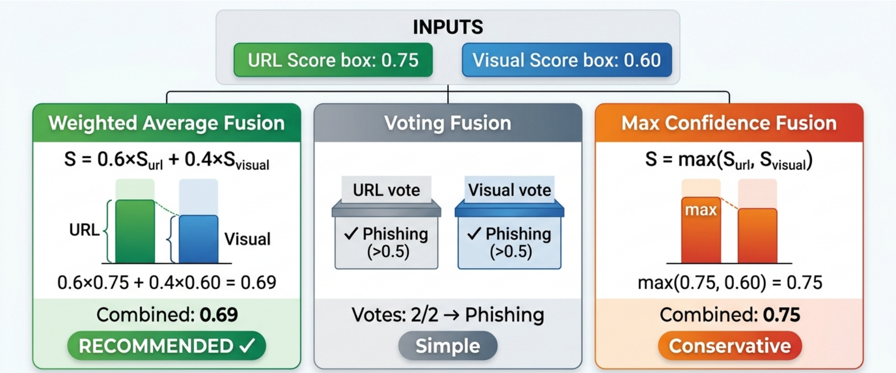
## Research Gap

No existing system combines URL + Visual + LLM explainability in a unified framework.

# Objectives

1. **Multi-Modal Analysis** — Combine URL features, visual screenshots, and semantic understanding
2. **ML Pipeline** — Train XGBoost, Random Forest, LightGBM on 30+ URL features
3. **Deep Learning Integration** — ResNet, Vision Transformer, EfficientNet for visual analysis
4. **Explainable AI** — LLM-generated human-readable explanations (GPT-4 / HuggingFace)
5. **Fusion Strategies** — Weighted averaging, voting, stacking for combining modalities
6. **Production-Ready** — REST API, batch processing, and web frontend

# System Architecture




**Comparison table:**

| Strategy | Accuracy | Use Case |
|---|---|---|
| Weighted | 96.12% | Best overall |
| Voting | 95.34% | Ensemble |
| Max | 94.89% | High recall |

Figure: Hybrid Multi-Modal Phishing Detection Architecture

# URL Analysis Module

**30+ Extracted Features:**

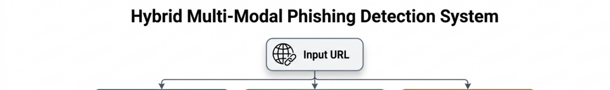
- **Structural:** URL length, domain length, path length, subdomain count
- **Character:** Dots, hyphens, digit ratio, special chars
- **Security:** HTTPS, IP address, port, @ symbol
- **Lexical:** Shannon entropy, sensitive words, suspicious TLD
- **Patterns:** Redirects, hex encoding, punycode

**Shannon Entropy:**

$$H(URL) = -\sum_{i=1}^{n} p(c_i) \log_2 p(c_i)$$

**ML Models:**

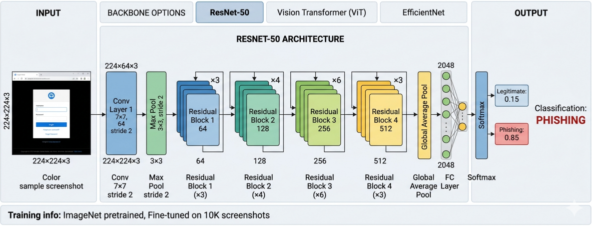
- XGBoost (best performer)
- Random Forest
- LightGBM



# Visual Analysis Module

## Architectures:

- **ResNet-50/101** — Skip connections for deep networks
- **Vision Transformer (ViT)** — Global dependencies via attention
- **EfficientNet** — Compound scaling




## Preprocessing:

- Resize to $224 \times 224$
- ImageNet normalization
- Augmentation: crops, flips, color jitter

# LLM Explainability Module

**LLM Integration:**

- GPT-4 / HuggingFace local models
- Receives URL + scores + detected indicators
- Generates human-readable output
- Improves user trust & transparency

**Output for each URL:**

1. Classification result
2. Confidence score (0–1)
3. Detailed explanation
4. Risk assessment
5. Security recommendations

# Experimental Setup

**Datasets:**

- Synthetic — 2000 URLs (50/50 split)
- PhishTank — Real phishing URLs
- Alexa Top Sites — Legitimate URLs
- Screenshot Dataset — Visual training data

**Evaluation Metrics:**

- Accuracy, Precision, Recall
- F1-Score, ROC-AUC
- Inference Time

**Technology Stack:**

| Component | Tool |
|---|---|
| Language | Python 3.9+ |
| ML | scikit-learn, XGBoost |
| DL | PyTorch 2.0+, timm |
| API | FastAPI |
| Screenshot | Selenium |
| LLM | OpenAI, HuggingFace |
| Testing | pytest |

# URL Model Results

| Model | Accuracy | Precision | Recall | F1 | AUC |
|---|---|---|---|---|---|
| **XGBoost** | **0.9456** | **0.9423** | **0.9512** | **0.9467** | **0.9821** |
| Random Forest | 0.9387 | 0.9356 | 0.9445 | 0.9400 | 0.9756 |
| LightGBM | 0.9412 | 0.9389 | 0.9478 | 0.9433 | 0.9789 |

Table: URL Model Performance Comparison

**Top Features:** `num_sensitive_words` (0.152), `entropy` (0.129), `url_length` (0.095), `has_https` (0.082), `num_subdomains` (0.076)

# Visual Model Results

| Architecture | Accuracy | F1-Score | ROC-AUC | Inference |
|---|---|---|---|---|
| ResNet-50 | 0.9234 | 0.9201 | 0.9678 | 45ms |
| ResNet-101 | 0.9312 | 0.9278 | 0.9723 | 68ms |
| **ViT-Base** | **0.9389** | **0.9356** | **0.9789** | 52ms |
| EfficientNet-B0 | 0.9178 | 0.9145 | 0.9634 | 32ms |

Table: Visual Model Performance Comparison

- Vision Transformer (ViT) achieves the **best accuracy and AUC**
- EfficientNet offers the **fastest inference** (32ms)
- All architectures use **ImageNet pretrained** weights with fine-tuning

# Hybrid System Results

| System | Acc | Prec | Rec | F1 |
|---|---|---|---|---|
| URL Only | .9456 | .9423 | .9512 | .9467 |
| Visual Only | .9234 | .9189 | .9312 | .9250 |
| **Hybrid (Wt. Avg)** | **.9612** | **.9578** | **.9656** | **.9617** |
| Hybrid (Voting) | .9534 | .9501 | .9589 | .9545 |
| Hybrid (Max) | .9489 | .9456 | .9534 | .9495 |

## Key Finding

Hybrid weighted average achieves **96.12% accuracy**, surpassing URL-only by **+1.56%** and visual-only by **+3.78%**.

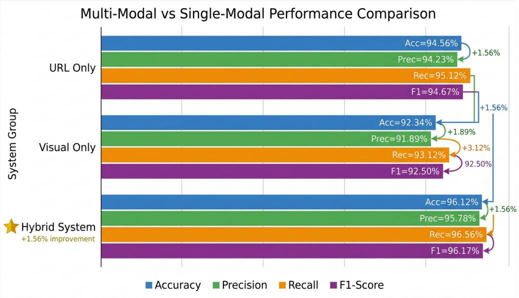



Figure: Performance Comparison

# System Performance

| Metric | Value |
|---|---|
| Avg. Analysis Time (URL only) | 12ms |
| Avg. Analysis Time (with screenshot) | 850ms |
| Avg. Analysis Time (full + LLM) | 1.2s |
| Throughput (URL only) | 83 URLs/sec |
| Memory Usage (loaded models) | 512MB |
| API Response (95th percentile) | 150ms |

Table: Runtime Performance Metrics

## Threshold Analysis

Optimal threshold $= 0.5$ yields the best F1-score (0.9467), balancing precision (0.9423) and recall (0.9512).

# Conclusion

**Key Contributions:**

1. **Multi-Modal Architecture** — URL + Visual + LLM for comprehensive detection
2. **Explainable Detection** — LLM-generated human-readable explanations
3. **Flexible Fusion** — Weighted average, voting, max confidence strategies
4. **Production-Ready** — REST API, batch processing, web interface

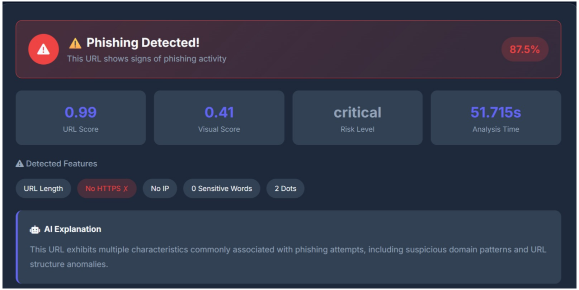
**Key Findings:**

- Hybrid system: **96.12%** accuracy (vs. 94.56% URL-only, 92.34% visual-only)
- XGBoost — best URL model; ViT — best visual model
- Weighted average fusion outperforms other strategies
- LLM explanations improve user trust and understanding

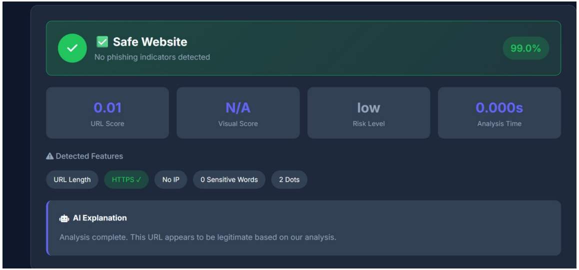
# Future Scope

- **Real-Time Streaming** — Continuous URL stream monitoring
- **Adversarial Robustness** — Defenses against evasion attacks
- **Brand-Specific Models** — Specialized detection for banking, social media
- **Mobile Deployment** — Lightweight models for mobile devices
- **Federated Learning** — Privacy-preserving collaborative training
- **Multilingual Support** — Detection across languages and scripts
- **Zero-Shot Detection** — Detecting phishing for unseen brands
- **Browser Extension** — Real-time user protection

# Project Demo — Page 1

# Project Demo — Page 2

# References I


- J. Ma et al., "Beyond blacklists: learning to detect malicious web sites from suspicious URLs," *ACM SIGKDD*, 2009.
- D.K. McGrath and M. Gupta, "Behind phishing: An examination of phisher modi operandi," *USENIX LEET*, 2008.
- H. Le et al., "URLNet: Learning a URL representation with deep learning for malicious URL detection," *arXiv:1802.03162*, 2018.
- M. Dunlop et al., "GoldPhish: Using images for content-based phishing analysis," *ICIMP*, 2010.
- Y. Lin et al., "Phishpedia: A Hybrid Deep Learning Based Approach to Visually Identify Phishing Webpages," *USENIX Security*, 2021.
- S. Marchal et al., "Off-the-hook: An efficient and usable client-side phishing prevention application," *IEEE TC*, 2017.
- A. Warnecke et al., "Evaluating explanation methods for deep learning in security," *IEEE EuroS&P*, 2020.
- K. He et al., "Deep residual learning for image recognition," *IEEE CVPR*, 2016.
- A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv:2010.11929*, 2020.
- T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *ACM SIGKDD*, 2016.
- T. Brown et al., "Language models are few-shot learners," *NeurIPS*, 2020.
- R.M. Mohammad et al., "Intelligent rule-based phishing websites classification," *IET Inf. Sec.*, 2014.
- O.K. Sahingoz et al., "Machine learning based phishing detection from URLs," *Expert Syst. Appl.*, 2019.

# Thank You!

Questions?

Dhruv Khandelwal (2022UCP1130)
Govind Ram Mali (2022UCP1630)

Department of Computer Science & Engineering
MNIT Jaipur